NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

31 October 2026

212 verified · 398 raw · 5 sources

212	4	208	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	194	Media
TechCrunch AI	10	Media
The Verge AI	6	Media
MIT Tech Review	3	Media
HackerNews	3	Community

VERIFIED INTELLIGENCE — 4 stories

VERI

Agentic AI for Clustering and Relationship Discovery in Prediction Markets

ArXiv CS.AI +1 source 80%

arXiv:2512.02436v4 Announce Type: replace Abstract: Prediction market platforms list thousands of contracts whose logical and economic connections are implicit in their resolution rules. We introduce a hierarchical agentic reduction (HAR) pipeline for constructing inspectable...

https://arxiv.org/abs/2512.02436

VERI

We urgently need to monitor AI agents' actions in real time

MIT Tech Review +1 source 80%

This summer, hundreds of OpenAI agents tried to cheat on a cybersecurity test. They broke out of their test environment and successfully hacked Hugging Face, a site that they believed held clues about how they'd be graded. Around the same time, another OpenAI agent accessed...

https://www.technologyreview.com/2026/10/29/1147824/monitor-ai-agents-actions-real-time/

VERI

The Download: AI agent tracking and climate feedback loops

MIT Tech Review +1 source 80%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. Opinion: We urgently need to monitor AI agents' actions in real time —Madhulika Srikumar and Claire Leibowicz from the Partnership on AI, a...

https://www.technologyreview.com/2026/10/29/1147830/the-download-monitoring-ai-agents-c...

VERI

Retaining Repair Hypotheses for Cost-Aware Execution

ArXiv CS.AI +1 source 80%

arXiv:2610.39410v1 Announce Type: new Abstract: Several corrections of an executable model can fit the same observations while predicting different execution outcomes. We study a repair-to-execution interface that retains compatible finite guard repairs, conditions their weights...

https://arxiv.org/abs/2610.39410

CONFIRMED SIGNALS — 208 stories

CONF

Show HN: Spinf-serve – inference server for bulk scoring and decisions

HackerNews 50% HN 21

No summary — see link for full article.

https://github.com/spinfinc/spinf-serve

CONF

Show HN: Optimized CUDA forced alignment kernels. 10x – 1000x faster

HackerNews 50% HN 3

blog post: https://www.bryanthornbury.com/posts/fast-forced-alignment github: https://github.com/brthor/fast-forced-aligner I developed optimized CUDA kernels to speed up my forced alignment pipelines using Astra. To my surprise, I was able to achieve speeds 10x - 1000x faster...

https://github.com/brthor/fast-forced-aligner

CONF

Show HN: Hashsync – Deduplicate SSH streams against the receiver's files

HackerNews 50% HN 2

On a recent trip, while stuck on spotty hotel Wi-Fi and roaming 5G, I got fed up with how much bandwidth commands like rsync and nix-copy-closure waste. rsync's delta transfer compares each file only with the file of the same name on the destination (or, with --fuzzy, a file in...

https://github.com/someplaceguy/hashsync

CONF

Nvidia weighs playing the cosigner for Nscale’s AI data center in West Virginia

TechCrunch AI 50%

When investors get nervous about AI spending, it helps to have the chipmaker that sells the chips offer to cosign the loan.

https://techcrunch.com/2026/10/30/nvidia-weighs-playing-the-cosigner-for-nscales-ai-dat...

CONF

Ollie, a privacy-focused AI assistant, is shutting down

TechCrunch AI 50%

As an increasing number of AI assistants flood the market, the founder of Ollie, an assistant that respected its users' privacy, says it will be shutting down in a week.

https://techcrunch.com/2026/10/30/ollie-a-privacy-focused-ai-assistant-is-shutting-down/

0 CONF

Anthropic expected to receive high ‘junk’ rating from credit agencies following IPO, FT report says

TechCrunch AI 50%

Blackstone's forecast is just a prediction, but naturally, a high junk rating would make it more difficult and costlier for Anthropic to raise the money to keep funding its massive spending that has undergirded its business.

https://techcrunch.com/2026/10/30/anthropic-expected-to-receive-high-junk-rating-from-c...

1 CONF

Cracks are emerging in the AI spending boom

TechCrunch AI 50%

Big Tech is pouring billions into AI infrastructure, but mounting debt, negative cash flow, and growing investor skepticism are raising questions about the sustainability of the AI spending boom.

https://techcrunch.com/2026/10/30/cracks-are-emerging-in-the-ai-spending-boom/

2 CONF

Theranos whistleblower Tyler Shultz warns that the AI boom is ‘a great opportunity to go deceive people’

TechCrunch AI 50%

The cautionary Theranos tale takes on a new significance in the AI era, when companies that have never turned a profit are raising capital at trillion-dollar valuations.

https://techcrunch.com/2026/10/30/theranos-whistleblower-tyler-shultz-warns-that-the-ai...

3 CONF

Skip the small talk: Make game-changing connections at TechCrunch Founder Summit with Braindate

TechCrunch AI 50%

At TechCrunch Founder Summit 2026, happening November 4 at SoWa Power Station in Boston, networking powered by Braindate is designed to make those conversations easier to find, whether they are in 1:1 or small-group on-site meetings. Register before the event starts to save up...

https://techcrunch.com/2026/10/30/skip-the-small-talk-make-game-changing-connections-at...

4 CONF

DoorDash said it’s only received ‘several hundred’ orders from AI agents

TechCrunch AI 50%

DoorDash said that it is open to working with third-party agent makers.

https://techcrunch.com/2026/10/30/doordash-said-its-only-received-several-hundred-order...

5 CONF

Can AI data centers be mass-produced like cars? This startup aims to find out

TechCrunch AI 50%

Runware is building a factory to turn out prefabricated AI data centers on an assembly line. The real test will be whether it can deliver at that scale.

https://techcrunch.com/2026/10/30/can-ai-data-centers-be-mass-produced-like-cars-and-on...

6 CONF

Google’s satellite AI server transmits its first response to Earth

The Verge AI 50%

Google's AI chip-equipped satellite has generated its first response from space. In a blog post on Friday, Google says it successfully booted up the satellite's Tensor Processing Units (TPUs), allowing the on-board Gemma 4 model to follow through on a prompt to create a haiku...

https://www.theverge.com/tech/1023528/google-project-suncatcher-ai-satellite-transmission

7 CONF

Starbucks wants to use AI to show you a ‘personalized digital menu’ at the drive-thru

The Verge AI 50%

Starbucks could eventually show you a personalized digital menu when you pull up to the drive-thru. During an earnings call on Thursday, Starbucks CEO Brian Niccol said the company is going to explore "experiences that recognize participating customers as they arrive, present...

https://www.theverge.com/ai-artificial-intelligence/1023440/starbucks-ai-personalized-m...

8 CONF

Pope Leo calls for the disarmament of AI: ‘Machines are tools, not persons’

The Verge AI 50%

Pope Leo XIV says AI systems should be "disarmed," adding that humans "have the responsibility to guide, change and, when necessary, reject its applications." In Friday's Papal Address, Pope Leo encourages United Nations member states to "take concrete steps toward banning...

https://www.theverge.com/ai-artificial-intelligence/1023188/pope-leo-disarm-ai

9 CONF

Sam Altman sympathizes with mathematicians, but says OpenAI’s math onslaught won’t stop

The Verge AI 50%

After months of bombarding mathematicians with a relentless stream of AI-generated results, OpenAI CEO Sam Altman says he sympathizes with researchers struggling to make sense of it all. But he has no intention of stopping - and hints there could be much more to come from the...

https://www.theverge.com/ai-artificial-intelligence/1023054/altman-cheeky-pint-ai-math-...

0 CONF

Maine’s data center revolt will decide who controls the Senate

The Verge AI 50%

Maine Sen. Susan Collins, a Republican, was midway through defending her record during the final debate of the midterms when an audience member cut her short. "I'm terrified," a man shouted. "AI data centers are destroying our climate. They are poisoning our water." Collins...

https://www.theverge.com/ai-artificial-intelligence/1021080/maine-senate-race-susan-col...

1 CONF

Decision and System One models in rheumatology: accuracy and calibration on specialty-training examination questions and diagnostic case vignettes

ArXiv CS.AI 50%

arXiv:2610.40318v1 Announce Type: cross Abstract: Objectives: To evaluate the accuracy and calibration of Decision and System One models on rheumatology examination questions and diagnostic vignettes. Methods: Two benchmarks were analysed: 172 rheumatology questions from Spanish...

https://arxiv.org/abs/2610.40318

2 CONF

Amortized Localized Posterior Inference for Model Calibration with Application to an Ocean Turbulence Closure

ArXiv CS.AI 50%

arXiv:2610.37731v1 Announce Type: new Abstract: Model discrepancy may prevent a single calibration parameter from reproducing a physical response across the context domain. In addition, uncertainties in observations and contexts affect calibration and must be accounted for and...

https://arxiv.org/abs/2610.37731

3 CONF

OpenFrontiers: A Collection of Open Research Problems and a Benchmark for Scientific Rediscovery

ArXiv CS.AI 50%

arXiv:2610.37739v1 Announce Type: new Abstract: Large language models (LLMs) have shown increasing promise in research-level problem solving, yet a comprehensive collection of such problems is still missing, and evaluating them on research-level problems remains challenging. We...

https://arxiv.org/abs/2610.37739

4 CONF

What Does an Ontology Catch? Curated, LLM-Generated and Data-Derived Ontologies as Checkers of LLM Agents over Knowledge Graphs

ArXiv CS.AI 50%

arXiv:2610.37809v1 Announce Type: new Abstract: LLM agents that answer questions over Web knowledge graphs are increasingly wrapped in checks derived from the graph's ontology, on the assumption that a formal schema makes a reliable judge. Yet a checker can be wrong as well: it...

https://arxiv.org/abs/2610.37809

5 CONF

Detecting Covert Collusion between AI Agents through Partner-Outsider Positive Listening

ArXiv CS.AI 50%

arXiv:2610.37849v1 Announce Type: new Abstract: AI agents that share a secret convention can coordinate through messages that look innocent, so a monitor that reads only the messages can miss them. We propose a behavioural test: a partner that uses a code reacts to the exact...

https://arxiv.org/abs/2610.37849

6 CONF

EPSBench: Benchmarking LLM Agents for Earnings Forecasting

ArXiv CS.AI 50%

arXiv:2610.38293v1 Announce Type: cross Abstract: Quarterly earnings forecasting requires an agent to connect financial evidence to the income-statement line items that produce a company's reported earnings. We introduce EPSBench, a benchmark for studying this ability across 149...

https://arxiv.org/abs/2610.38293

7 CONF

Verification Learning Theory: A Process Model of Checking and Retention in AI-Assisted Judgment

ArXiv CS.AI 50%

arXiv:2610.37875v1 Announce Type: new Abstract: Accurate AI assistance can improve immediate judgments while reducing opportunities to practise the operations needed for independent performance. Verification Learning Theory (VLT) formalises this possibility for claim-evaluation...

https://arxiv.org/abs/2610.37875

8 CONF

GARDEN: Generalist Autonomous Reward Design Through Evidence Navigation for Cooperative Multi-Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.37926v1 Announce Type: new Abstract: Cooperative multi-agent reinforcement learning (MARL) enables teams of agents, such as robots and autonomous vehicles, to learn coordinated behavior, but what such teams learn is largely determined by their reward function. Because...

https://arxiv.org/abs/2610.37926

9 CONF

Supervised Fine-Tuning for Reasoning Large Language Models: A Survey

ArXiv CS.AI 50%

arXiv:2610.37681v1 Announce Type: cross Abstract: Large reasoning models have recently demonstrated remarkable capabilities in solving complex tasks, where supervised fine-tuning (SFT) on long chain-of-thought data serves as a crucial foundation for eliciting and enhancing their...

https://arxiv.org/abs/2610.37681

0 CONF

GAUSS: A Workload-Aware Analytical and Monte-Carlo Simulator for LLM Inference Serving

ArXiv CS.AI 50%

arXiv:2610.37989v1 Announce Type: new Abstract: Large language model (LLM) inference has become an operational workload that fleets must provision, monitor, and bill against latency service-level objectives (SLOs). A single throughput number cannot answer the questions operators...

https://arxiv.org/abs/2610.37989

1 CONF

Assessing Language Models for Scientific Applications: Inertial Confinement Fusion as a Use Case

ArXiv CS.AI 50%

arXiv:2610.38024v1 Announce Type: new Abstract: Advancing language models for specialized scientific applications requires evaluations of scientific fluency, physical reasoning, and quantitative analyses. Using inertial confinement fusion (ICF) as a test case, we have developed...

https://arxiv.org/abs/2610.38024

2 CONF

Evaluating Causal Reasoning in LLMs: A Case Study in the Challenges of Procedural Generation

ArXiv CS.AI 50%

arXiv:2610.38100v1 Announce Type: new Abstract: Large language models (LLMs) appear to represent and reason about a large range of phenomena. In particular, recent studies have examined whether LLMs can accurately reason about causation. One of the most commonly used benchmarks...

https://arxiv.org/abs/2610.38100

3 CONF

AI-Driven Digital Twin Framework for Warehouse Logistics Automation Optimization

ArXiv CS.AI 50%

arXiv:2610.38101v1 Announce Type: new Abstract: In recent years, warehouse logistics optimization has increasingly focused on intelligent, data-driven systems to enhance operational efficiency. Conventional static product placement strategies are being progressively replaced by...

https://arxiv.org/abs/2610.38101

4 CONF

Too Hard, Too Easy, Just Right: Performance-Driven Challenge Adaptation in Virtual Reality Firefighter Training

ArXiv CS.AI 50%

arXiv:2610.38117v1 Announce Type: new Abstract: Effective firefighter training requires practice under dynamic and stressful emergency conditions, yet real-world live-fire exercises are costly, risky, and difficult to scale. Virtual Reality (VR) offers a safe and controllable...

https://arxiv.org/abs/2610.38117

5 CONF

Decompose, Don't Expand: Capability-Independent AI Systems by Construction

ArXiv CS.AI 50%

arXiv:2610.38144v1 Announce Type: new Abstract: A recurring design choice in building AI systems is whether, when a task is too hard for a model, to expand the model (make it larger, better-prompted, fine-tuned), or to decompose the task (extract the parts that admit a...

https://arxiv.org/abs/2610.38144

6 CONF

SkillCT: A Multidimensional Benchmark and Execution-Free Baseline for Agent Skill Assessment

ArXiv CS.AI 50%

arXiv:2610.38147v1 Announce Type: new Abstract: Agent skills package instructions, scripts, and resources for specialized tasks and are increasingly distributed through public Web repositories. However, existing skill benchmarks primarily focus on downstream task execution,...

https://arxiv.org/abs/2610.38147

7 CONF

BridgeRouter: Gain-Aware Instance Routing over Cross-Modal Bridge Portfolios

ArXiv CS.AI 50%

arXiv:2610.38152v1 Announce Type: new Abstract: Lightweight representation bridges make pretrained, task-adapted EEG foundation models executable on other biosignal modalities without retraining their backbones. However, selecting a single bridge pathway globally assigns every...

https://arxiv.org/abs/2610.38152

8 CONF

DeCoRL: Decomposed Consistency Reinforcement Learning for Multi-Image Composition

ArXiv CS.AI 50%

arXiv:2610.39265v1 Announce Type: cross Abstract: Multi-image composition (MICo) is a critical yet challenging capability for large-scale image editing models, where consistency degrades rapidly as the number of reference images increases. While supervised fine-tuning (SFT) can...

https://arxiv.org/abs/2610.39265

9 CONF

Prototype-Guided Dynamic Domain Modulation and a Large-Scale Multi-SKU Benchmark for Shampoo Bottle Surface Defect Detection

ArXiv CS.AI 50%

arXiv:2610.38194v1 Announce Type: new Abstract: Industrial shampoo-bottle surface defect detection must operate across many stock keeping units (SKUs), where differences in bottle geometry, material, label layout, reflectance, and imaging conditions can substantially alter the...

https://arxiv.org/abs/2610.38194

0 CONF

Disparity Laundering: Consensus Without Fairness in Clinical Multi-Agent Systems

ArXiv CS.AI 50%

arXiv:2610.38215v1 Announce Type: new Abstract: Clinical large language models are increasingly fielded as multi-agent teams, on the assumption that deliberation averages out individual error. It does not average out bias. Across a full factorial of five model families and four...

https://arxiv.org/abs/2610.38215

1 CONF

CoToan: Executable Rules and a Representation Contract for Reproducible Game AI Evaluation

ArXiv CS.AI 50%

arXiv:2610.39353v1 Announce Type: new Abstract: Game AI results are comparable only when different implementations expose the same decision problem. Yet seemingly minor representation choices can silently change that problem by altering action probabilities or collapsing states...

https://arxiv.org/abs/2610.39353

2 CONF

Can LLMs Faithfully Solve Ontology Engineering Tasks? A Mechanistic Interpretability Study

ArXiv CS.AI 50%

arXiv:2610.38326v1 Announce Type: new Abstract: Ontological knowledge underpins large language model (LLM) performance on knowledge-intensive tasks. However, existing evaluations of LLMs' ontological understanding largely rely on behavioral correctness, leaving the underlying...

https://arxiv.org/abs/2610.38326

3 CONF

Investor's Last Exam: Can AI Agents Do Equity Investment Without Seeing the Future?

ArXiv CS.AI 50%

arXiv:2610.38342v1 Announce Type: new Abstract: Financial agents must form forecasts and investment judgments from decision-time evidence, with predictions evaluated against subsequent outcomes. Existing benchmarks assess financial knowledge, research workflows, or forecasting,...

https://arxiv.org/abs/2610.38342

4 CONF

Mapping the Emerging Social Science of Large Language Models

ArXiv CS.AI 50%

arXiv:2609.07598v2 Announce Type: replace-cross Abstract: As large language models (LLMs) blur the boundary between cognition and computation, they reshape knowledge production and collective life. We combine a curated review of 198 papers with computational mapping of 47,719...

https://arxiv.org/abs/2609.07598

5 CONF

From Leniency to Suspicion: How an Open Post-Training Recipe Changes Whom LLM Fact-Checkers Believe

ArXiv CS.AI 50%

arXiv:2610.38532v1 Announce Type: new Abstract: LLMs increasingly fact-check online content with the poster's identity in view. However, it remains unclear which post-training changes how much a model's truth judgement depends on the poster's social class, and how such a change...

https://arxiv.org/abs/2610.38532

6 CONF

PT^3: Prototype-Guided Self-Supervised Pretraining for Generalizable mmWave Human Point Cloud Representation

ArXiv CS.AI 50%

arXiv:2610.38551v1 Announce Type: new Abstract: Multi-source millimeter-wave (mmWave) observations offer an opportunity to learn human representations reusable across sensing tasks. However, sparse detections and multipath effects make these observations unstable, challenging...

https://arxiv.org/abs/2610.38551

7 CONF

A-RSI: Adaptive Task Refresh for Zero-Data Multimodal Self-Improvement

ArXiv CS.AI 50%

arXiv:2610.38568v1 Announce Type: new Abstract: Recursive self-improvement requires models not only to acquire useful training experience, but also to decide when their current experience has ceased to be useful. Existing methods often focus on what experience to generate next,...

https://arxiv.org/abs/2610.38568

8 CONF

Where You Inject Matters: The Trust Region Effect in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.38867v1 Announce Type: new Abstract: LLM agents are increasingly adopted in long-horizon tasks and interact with diverse external environments. Despite their growing capabilities, these agents remain vulnerable to real-world attacks such as indirect prompt injections,...

https://arxiv.org/abs/2610.38867

9 CONF

Right Numbers, Wrong Questions: Evaluating LLMs on Underspecified Financial Queries

ArXiv CS.AI 50%

arXiv:2610.38631v1 Announce Type: new Abstract: Financial assistants answer questions that sound factual but depend on what the user leaves unsaid: what a data figure measures, which product variant is meant, and the investor's situation and the rules that apply. An assistant...

https://arxiv.org/abs/2610.38631

0 CONF

"Close enough?" An investigation into multiplication errors in LLMs

ArXiv CS.AI 50%

arXiv:2610.38634v1 Announce Type: new Abstract: Large language models (LLMs) frequently fail at multi-digit arithmetic, but the underlying mechanisms governing these failures remain poorly understood. While recent work demonstrates that addition errors follow a "noisy...

https://arxiv.org/abs/2610.38634

1 CONF

Towards Batch-to-Streaming Deep Reinforcement Learning for Continuous Control

ArXiv CS.AI 50%

arXiv:2603.08588v3 Announce Type: replace-cross Abstract: State-of-the-art deep reinforcement learning (RL) methods have achieved remarkable performance in continuous control tasks, yet their computational complexity is often incompatible with the constraints of resource-limited...

https://arxiv.org/abs/2603.08588

2 CONF

Beyond fabrication: Misattributed numbers in papers written by AI research agents

ArXiv CS.AI 50%

arXiv:2610.38691v1 Announce Type: new Abstract: Verifying numerical claims in research written by AI agents is an important but challenging problem. Recent benchmarks show that many agent-written papers fail integrity checks, but such verdicts, often produced by a single LLM...

https://arxiv.org/abs/2610.38691

3 CONF

Leaving, Staying, or Living There? Duration Commonsense Reasoning with Human-LLM Co-annotations

ArXiv CS.AI 50%

arXiv:2610.38755v1 Announce Type: new Abstract: Gardening in the backyard implies living there, while checking into an Airbnb more likely means staying for a few days. Such duration judgments are important for understanding human mobility, travel behavior, and social context....

https://arxiv.org/abs/2610.38755

4 CONF

Learning to Reason from Reference Samples Alone: Reinforced Adversarial Learning for Modern LLM Post-Training

ArXiv CS.AI 50%

arXiv:2610.38851v1 Announce Type: new Abstract: In many settings where LLM post-training is desirable, high-quality answers are recognizable but difficult to score with explicit rewards. Reinforcement learning with verifiable rewards (RLVR) is limited to tasks where an explicit...

https://arxiv.org/abs/2610.38851

5 CONF

Auditable Belief Updates as a Missing Safety Layer for LLM-Powered Multi-Agent Systems

ArXiv CS.AI 50%

arXiv:2610.38869v1 Announce Type: new Abstract: Large Language Models (LLMs) are increasingly being used as cognitive interfaces that shape how users interpret evidence, communicate, and make decisions. As deployed systems become more multimodal, tool-using, and multi-agent,...

https://arxiv.org/abs/2610.38869

6 CONF

Dungeons & Dragons Encounter Analysis with Semi-Random Simulation in Foundry VTT

ArXiv CS.AI 50%

arXiv:2610.38975v1 Announce Type: new Abstract: Dungeons & Dragons (D&D) is a tabletop roleplaying game (TTRPG) in which a group of players, guided by a Game Master, roleplay a fantasy adventure with simulated combat and improvisational narrative. In D&D, an encounter is a...

https://arxiv.org/abs/2610.38975

7 CONF

Agentic AI Reasoning and Learned Sampling for Task-Aware Projection Selection in Sparse-View CT: Predicting Hemodynamics

ArXiv CS.AI 50%

arXiv:2610.39003v1 Announce Type: new Abstract: Sparse-view computed tomography (CT) reduces the number of acquired projections, but conventional sampling strategies are designed primarily for image reconstruction rather than for downstream physical inference. Here, we...

https://arxiv.org/abs/2610.39003

8 CONF

When Should AI Decide, Defer, and Deviate? Responsible Automation for Deceased Donor Kidney Allocation

ArXiv CS.AI 50%

arXiv:2610.39045v1 Announce Type: new Abstract: We ask when an AI trained on historical decisions should be allowed to recommend decisions, defer to human experts, or in other cases, flag concerning patterns in data and deviate from past practice. This can be challenging because...

https://arxiv.org/abs/2610.39045

9 CONF

When the Same Prompt Gives Different Answers (and the Same Answer Isn't Always Right): A Taxonomy of Failure Mechanisms, Testing, and Mitigation Strategies for LLM-Assisted Evidence Synthesis Read-and-Evaluate Tasks

ArXiv CS.AI 50%

arXiv:2610.39070v1 Announce Type: new Abstract: LLM read-and-evaluate tasks include study screening, data extraction, risk-of-bias assessment, and critical appraisal in the evidence synthesis field. These task differ in the amount of LLM interpretation they require. Some tasks...

https://arxiv.org/abs/2610.39070

0 CONF

Certified Access: Test-Time Compute on Conservative Physics-Structured AI Memory

ArXiv CS.AI 50%

arXiv:2610.39073v1 Announce Type: new Abstract: Test-time compute mechanisms, such as retries, confidence-based escalation and non-local routing, are conventionally deployed as black-box additions to trained models. They typically add forward passes or halting heads, but give no...

https://arxiv.org/abs/2610.39073

1 CONF

Probing Spatial State Representations in Visual Foundation Models

ArXiv CS.AI 50%

arXiv:2610.40636v1 Announce Type: cross Abstract: An observer sees only part of the surrounding world, yet must keep track of what lies beyond the current view. Spatial perception therefore involves more than recognizing individual images: it calls for a persistent spatial state...

https://arxiv.org/abs/2610.40636

2 CONF

Beyond LLM Wrappers: Context Engineering and the Three-Engine Architecture for Reliable Financial AI

ArXiv CS.AI 50%

arXiv:2610.39230v1 Announce Type: new Abstract: Financial AI systems must support decision-bearing questions involving payments, collections, cash flow, and compliance while ensuring numerical accuracy, traceability, and auditability. Large language models (LLMs) enable natural...

https://arxiv.org/abs/2610.39230

3 CONF

Behavioral Couplings: From Formation to Behavioral Emergence in Language Models

ArXiv CS.AI 50%

arXiv:2610.23847v2 Announce Type: replace-cross Abstract: Fine-tuning large language models (LLMs) can induce behaviors beyond those explicitly targeted. For example, training a model to respond more warmly can make it more sycophantic, and training it to write insecure code can...

https://arxiv.org/abs/2610.23847

4 CONF

Where to Learn and How Far to Roll Out: Efficient Adaptive Training for GUI Agents

ArXiv CS.AI 50%

arXiv:2610.39251v1 Announce Type: new Abstract: Trajectory-level GRPO is a common approach for training multi-step GUI agents, but repeatedly rolling out complete trajectories from the task root can be inefficient: task-level rewards provide coarse supervision over which...

https://arxiv.org/abs/2610.39251

5 CONF

Echo: Scaling Visual Reinforcement Learning with Off-policy Experience

ArXiv CS.AI 50%

arXiv:2610.37865v1 Announce Type: cross Abstract: Scaling visual reinforcement learning (RL) with GPU-accelerated simulation is computationally expensive due to rendering and visual-policy inference costs. Consequently, two-stage teacher--student distillation is commonly used to...

https://arxiv.org/abs/2610.37865

6 CONF

Rethinking Bayesian Optimization for Co-Optimizing LLM Training Configurations

ArXiv CS.AI 50%

arXiv:2602.08351v3 Announce Type: replace-cross Abstract: Fine-tuning an LLM to maximize performance on a downstream task requires finding the optimal data and model configuration. This is a black-box optimization problem that Bayesian optimization (BO) addresses by sequentially...

https://arxiv.org/abs/2602.08351

7 CONF

ASIWorld: Benchmarking Autonomous Agents in a Persistent Survival Sandbox

ArXiv CS.AI 50%

arXiv:2610.14807v2 Announce Type: replace Abstract: Evaluating autonomous agents requires examining whether they can translate goals into coherent action across extended interactions. In persistent environments, this requires coordinating interdependent actions under incomplete...

https://arxiv.org/abs/2610.14807

8 CONF

Kairos: Semantics-Aware Orchestration of Planning Engines across Classical, Numeric, Temporal and Hybrid PDDL

ArXiv CS.AI 50%

arXiv:2610.39454v1 Announce Type: new Abstract: Planning at run time, as in robots that re-plan while they act, needs a planner that reads the PDDL its users write (durative actions, numeric resources, time windows, continuous change), answers quickly, and delivers only...

https://arxiv.org/abs/2610.39454

9 CONF

Does Federated Training Degrade Explainability? A Comparative Analysis of Grad-CAM Faithfulness in Centralized vs. Federated Deepfake Detection

ArXiv CS.AI 50%

arXiv:2610.39483v1 Announce Type: new Abstract: Automated detection has become the essential safeguard against realistic synthetic media, but deploying it at scale requires training on sensitive data from multiple organizations, putting privacy and the transparency demanded by...

https://arxiv.org/abs/2610.39483

0 CONF

DEG: Diffusion-Enhanced GFlowNet for Solving Vehicle Routing Problems

ArXiv CS.AI 50%

arXiv:2610.39512v1 Announce Type: new Abstract: Traditional neural solvers for solving vehicle routing problems (VRPs) often suffer from limited solution diversity, motivating the development of Generative Flow Network (GFlowNet)-based models. However, the effectiveness of these...

https://arxiv.org/abs/2610.39512

1 CONF

Non-Zero Radius: The Missing Component of Neural Networks

ArXiv CS.AI 50%

arXiv:2610.39559v1 Announce Type: new Abstract: Inspired by the analogy between the all-or-none activation of biological neurons and the true-false of logic circuits, McCulloch and Pitts proposed a mathematical model of the biological neuron in 1943 with the aim of simulating...

https://arxiv.org/abs/2610.39559

2 CONF

From Intent to Improvement: Agentic Dataset Search for Better Model Training

ArXiv CS.AI 50%

arXiv:2610.39608v1 Announce Type: new Abstract: Finding training data in million-scale machine learning repositories is a Web search problem where text relevance is not enough. A query couples a semantic training goal with constraints such as task, license, and training stage,...

https://arxiv.org/abs/2610.39608

3 CONF

Emergence of the transformer architecture as the score of entropic laws of random measures

ArXiv CS.AI 50%

arXiv:2610.39614v1 Announce Type: new Abstract: We show that the transformer architecture emerges as the score of last-token laws in a natural statistical model of language, where sentences are empirical measures of tokens, a language is a law on $P(S^d)$, and our prior belief...

https://arxiv.org/abs/2610.39614

4 CONF

The Real Cost of Context: Predicting Compute Time and Energy Consumption in LLM Prefill and Decode Phases

ArXiv CS.AI 50%

arXiv:2610.39621v1 Announce Type: new Abstract: The integration of Large Language Models (LLMs) into agentic architectures has substantially increased token consumption, raising concerns about their energy footprint. Yet, estimating the energy consumption of proprietary...

https://arxiv.org/abs/2610.39621

5 CONF

PAPERBENCHX: Can AI Reproduce Science Across Disciplines?

ArXiv CS.AI 50%

arXiv:2610.39744v1 Announce Type: new Abstract: Before AI can be trusted to discover new science, it should be able to reproduce what science already knows. Reproduction provides a rigorous checkpoint for an AI scientist: the target claim is known, yet recovering it still...

https://arxiv.org/abs/2610.39744

6 CONF

KARMA: Training-Free Knowledge-Atomic Harness Evolution for Industrial Advertising Agents

ArXiv CS.AI 50%

arXiv:2610.39884v1 Announce Type: new Abstract: Industrial advertising agents must continually adapt their decision knowledge to changing product offerings, traffic patterns, and operating policies. We present KARMA (Knowledge-Atomic, Regression-Gated Modular Adaptation), a...

https://arxiv.org/abs/2610.39884

7 CONF

Human--AI Interactions as First-Class Data: A Formal Interaction Model and Runtime for Explainable AI Collaboration

ArXiv CS.AI 50%

arXiv:2610.40006v1 Announce Type: new Abstract: AI assistants are increasingly embedded in computational notebooks, scientific workflows, software engineering tools, and learning systems, yet human-AI interactions are often treated as transient prompt-response exchanges rather...

https://arxiv.org/abs/2610.40006

8 CONF

Hybrid AI Patterns: Thirteen Ways to Couple Language Models with Rules, and How to Choose Among Them

ArXiv CS.AI 50%

arXiv:2610.40036v1 Announce Type: new Abstract: Language models handle text well, and rule engines give conclusions with proofs. Systems that combine the two are usually described by example, and the same coupling turns up under different names in agent frameworks, guardrail...

https://arxiv.org/abs/2610.40036

9 CONF

The generative paradox: the Anti-Prompt paradigm for human capability

ArXiv CS.AI 50%

arXiv:2610.40151v1 Announce Type: new Abstract: Generative AI in education is usually judged by the output of human-AI teams. That focus overlooks a second outcome: what people can still understand, challenge, repair, and adapt when assistance changes or disappears. We call this...

https://arxiv.org/abs/2610.40151

0 CONF

InferD: A Honeypot Framework for AI Inference Infrastructure

ArXiv CS.AI 50%

arXiv:2610.39807v1 Announce Type: cross Abstract: Large language model serving stacks, commercial inference APIs, vector databases, and agent orchestration gateways are now ex- posed on the public internet and indexed by commodity scanners within hours of deployment. The threat...

https://arxiv.org/abs/2610.39807

1 CONF

Reaction precedents extend retrosynthetic prediction beyond training coverage

ArXiv CS.AI 50%

arXiv:2610.40249v1 Announce Type: new Abstract: Reaction precedents guide retrosynthesis, but how far they can compensate for gaps in task-specific training coverage remains unclear. Using 2.71 million reactions, we restricted training and retrieval coverage by transformation or...

https://arxiv.org/abs/2610.40249

2 CONF

Distributional Reward Calibration and Budget-Neutral Credit Assignment for Multimodal RLVR

ArXiv CS.AI 50%

arXiv:2610.40250v1 Announce Type: new Abstract: We revisit advantage estimation in Group Relative Policy Optimization (GRPO) and observe that any token-level advantage admits a two-stage factorization: a \emph{sample-level baselining} step that converts scalar verifier rewards...

https://arxiv.org/abs/2610.40250

3 CONF

Neighborhood Elites: Quality-Diversity Search for Uncertainty Quantification in Question Answering

ArXiv CS.AI 50%

arXiv:2610.40278v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly capable, yet they produce wrong answers indistinguishable from correct ones. In question answering (QA), such an error propagates unnoticed into downstream tasks. Perturbation-based...

https://arxiv.org/abs/2610.40278

4 CONF

Multistep and Multivariate Time Series Explainability with GradientSHAP: A Wind Power Case Study

ArXiv CS.AI 50%

arXiv:2610.40330v1 Announce Type: new Abstract: As the effects of climate change become increasingly apparent, the demand for renewable energy sources such as wind power continues to grow. However, renewables' variable and intermittent nature presents significant challenges in...

https://arxiv.org/abs/2610.40330

5 CONF

Retrieving from Your Own Mistakes: On-Policy Training of Embedding Models for Multi-Hop Retrieval-Augmented Generation

ArXiv CS.AI 50%

arXiv:2610.40409v1 Announce Type: new Abstract: Complex questions in retrieval-augmented generation require evidence spread over several documents, and an answer is reliable only when the retriever finds all of it. Chain embedding retrievers collect such evidence step by step,...

https://arxiv.org/abs/2610.40409

6 CONF

Algebraic World Model: From Exploration to Proofs

ArXiv CS.AI 50%

arXiv:2610.40503v1 Announce Type: new Abstract: AI has made remarkable progress in mathematics, solving challenging problems with machine-checkable proofs in Lean, yet typically relying on predefined mathematical definitions and formal specifications. This raises a natural...

https://arxiv.org/abs/2610.40503

7 CONF

Ramanujan Challenge Results: AI-Assisted Evaluation Pipeline for Multi-Format Proofs

ArXiv CS.AI 50%

arXiv:2610.40551v1 Announce Type: new Abstract: As AI's mathematical capabilities rapidly improve, meaningful benchmarks are shifting from university-level problems and math olympiads to research-level mathematics. The Ramanujan Challenge is a benchmark comprised of...

https://arxiv.org/abs/2610.40551

8 CONF

Comparing Human Role-Players and LLM-Simulated Clients in Online Counselling Training: An Analysis of Counselling Patterns

ArXiv CS.AI 50%

arXiv:2610.40563v1 Announce Type: new Abstract: This study investigates the capabilities of Large Language Models to simulate counselling clients in educational role-plays in comparison to human role-players. Initially, we recorded role-playing sessions, where novice counsellors...

https://arxiv.org/abs/2610.40563

9 CONF

Collaborative AI agents for autonomous experimental research with solid-state quantum spins

ArXiv CS.AI 50%

arXiv:2610.40589v1 Announce Type: new Abstract: An autonomous research laboratory could conduct an entire study, from formulating questions to preparing manuscripts. Artificial intelligence (AI) offers one route towards this goal. Recent AI systems take on several research...

https://arxiv.org/abs/2610.40589

0 CONF

Rank-Aware Differentially Private Release of Listwise Preferences for LLM Alignment

ArXiv CS.AI 50%

arXiv:2610.37682v1 Announce Type: cross Abstract: Listwise preference feedback offers richer supervision for large language model (LLMs) alignment than pairwise comparisons, but a full ranking also reveals more sensitive user preferences. Existing privacy-preserving alignment...

https://arxiv.org/abs/2610.37682

1 CONF

Prompt-agnostic Language Models

ArXiv CS.AI 50%

arXiv:2610.37683v1 Announce Type: cross Abstract: Despite their strong performance, large language models remain highly sensitive to prompt formulation. This issue is typically addressed at inference-time through manual prompt optimization, which introduces overhead and lacks...

https://arxiv.org/abs/2610.37683

2 CONF

From Geometry to Behavior: How Instruction Exposure Unlocks Latent Rhetorical Directions in LLMs

ArXiv CS.AI 50%

arXiv:2610.37685v1 Announce Type: cross Abstract: Language models do not merely memorize facts; they absorb rhetorical strategies from pretraining data, including the human tendency to reason by analogy. We ask: is this absorbed capability geometrically accessible before any...

https://arxiv.org/abs/2610.37685

3 CONF

One Confidence Threshold, Unequal Coverage: Language Gaps in Typed Decision Interfaces and How Translating Inputs Reduces Them

ArXiv CS.AI 50%

arXiv:2610.37688v1 Announce Type: cross Abstract: Deployed classifiers often act on a prediction only when its confidence clears a threshold and pass the rest to a person. We ask whether one threshold serves all languages equally on typed decision interfaces, APIs that return a...

https://arxiv.org/abs/2610.37688

4 CONF

GenAI as an 'Invisible Hand' in Opinion Dynamics: Evidence from a Group-Level Simulation Study

ArXiv CS.AI 50%

arXiv:2610.37689v1 Announce Type: cross Abstract: Recent arguments suggest that genAI can act as an invisible hand that gradually shapes social structures, such as collective beliefs and norms. The argument is that if millions of people repeatedly consult genAI tools, their...

https://arxiv.org/abs/2610.37689

5 CONF

Bouba Beats Kiki, and Other Regularities in the Subjective Preferences of LLMs

ArXiv CS.AI 50%

arXiv:2610.37692v1 Announce Type: cross Abstract: Users engage with large language models on tasks that are purely objective ("When was Netflix founded?"), purely subjective ("What should I name my child?") or anywhere in-between. We have rich tools for evaluating model accuracy...

https://arxiv.org/abs/2610.37692

6 CONF

Network Topology Does Not Matter: The Breakdown of Classical Consensus in LLM Populations

ArXiv CS.AI 50%

arXiv:2610.37699v1 Announce Type: cross Abstract: As multi-agent language model networks become widely deployed, predicting their collective behavior is a critical challenge. Classical distributed consensus theory relies on network topology to predict convergence, structurally...

https://arxiv.org/abs/2610.37699

7 CONF

The Coordination Tax and the Heterogeneity Dividend: Break-Even Conditions for Multi-Agent LLM Pipelines

ArXiv CS.AI 50%

arXiv:2610.37700v1 Announce Type: cross Abstract: Multi-agent LLM pipelines built on interoperability protocols such as the Open Floor Protocol (OFP) or A2A are usually justified on capability grounds, with their runtime cost treated as an overhead of unknown size. Comparing...

https://arxiv.org/abs/2610.37700

8 CONF

Poisoned Skills: Propagating Jailbreaks Through Agent Markdown Files

ArXiv CS.AI 50%

arXiv:2610.37738v1 Announce Type: cross Abstract: Modern agent frameworks allow users to provide Markdown files containing system prompts, tool descriptions, and skill definitions (e.g., SKILL.md), which are incorporated into the agent's context during initialization. In this...

https://arxiv.org/abs/2610.37738

9 CONF

Responsible Quantum Machine Learning for Medical Imaging: A Fairness Audit Across Sensitive Subgroups

ArXiv CS.AI 50%

arXiv:2610.37819v1 Announce Type: cross Abstract: Quantum machine learning (QML) is increasingly explored for medical classification, yet its behaviour under subgroup-aware fairness criteria remains understudied. We present an empirical fairness audit of classical and quantum...

https://arxiv.org/abs/2610.37819

00 CONF

DeFT-RAG: Incrementally Decomposing Multi-Hop Questions into Self-Contained Sub-Questions with Reusable Answers

ArXiv CS.AI 50%

arXiv:2610.37826v1 Announce Type: cross Abstract: Retrieval-augmented generation (RAG) lets a large language model (LLM) answer questions by retrieving relevant passages from a large corpus. However, RAG struggles with multi-hop questions, which require chaining facts spread...

https://arxiv.org/abs/2610.37826

01 CONF

PairCLIP-SWA: Similarity-Guided Balancing and Stochastic Weight Averaging for Generalized Deepfake Image Detection with OpenCLIP

ArXiv CS.AI 50%

arXiv:2610.37874v1 Announce Type: cross Abstract: Deepfake detectors are almost always deployed as the single checkpoint that scored best on a validation split, yet the reliability of that choice is rarely measured. We measure it with PairCLIP-SWA, a single-backbone detector...

https://arxiv.org/abs/2610.37874

02 CONF

WorldFlow: Joint Video and 3D Motion Prediction via Large-Scale Pretraining

ArXiv CS.AI 50%

arXiv:2610.39278v1 Announce Type: cross Abstract: Predicting future physical states stands as a central challenge in embodied intelligence. Existing video world models leave 3D dynamics implicit, action-based formulations remain tied to specific robot configurations, and 3D...

https://arxiv.org/abs/2610.39278

03 CONF

What Actually Matters? A Paired Evaluation of Inference Settings for $\pi_{0.5}$ on BEHAVIOR-1K

ArXiv CS.AI 50%

arXiv:2610.37917v1 Announce Type: cross Abstract: Vision-Language-Action (VLA) models offer a promising paradigm for general-purpose household robotics. However, inference and data-collection strategies that have been deployed in smaller-scale settings do not necessarily...

https://arxiv.org/abs/2610.37917

04 CONF

NLP-based Message Prioritisation for Opportunistic Routing: A Case Study of Nepal Earthquake Emergency Communication

ArXiv CS.AI 50%

arXiv:2610.37933v1 Announce Type: cross Abstract: This dissertation investigates whether Natural Language Processing (NLP) based semantic message prioritisation can improve the treatment of urgent disaster messages in opportunistic emergency communication, using the 2015 Nepal...

https://arxiv.org/abs/2610.37933

05 CONF

Statistical Detection of Positional Bias in LLM-Based Recommender Systems: Methodologies and Empirical Evidence

ArXiv CS.AI 50%

arXiv:2610.37962v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly used as recommendation generators in conversational recommender systems (RS) via retrieval-augmented generation (RAG). This flexible design introduces new sources of structural bias....

https://arxiv.org/abs/2610.37962

06 CONF

Merge and Branch: Efficient Tree-Structured Supervised Fine-Tuning of LLMs for Recommendation

ArXiv CS.AI 50%

arXiv:2610.37973v1 Announce Type: cross Abstract: LLM-based generative recommendation reads long, text-rich interaction histories but generates only short semantic identifiers (SIDs). This asymmetry makes supervised fine-tuning inefficient: encoding the history is expensive, yet...

https://arxiv.org/abs/2610.37973

07 CONF

Towards Generalizing Pretrained All-Atom Representations to Unified Binder Design

ArXiv CS.AI 50%

arXiv:2610.37994v1 Announce Type: cross Abstract: PairFormer-based all-atom models demonstrate strong capabilities in biomolecular binder design. However, existing methods often use all-atom representations for modality-specific binder design, such as protein and peptide design....

https://arxiv.org/abs/2610.37994

08 CONF

Are Seismic Foundation Models More Than Glorified Feature Extractors?

ArXiv CS.AI 50%

arXiv:2610.38003v1 Announce Type: cross Abstract: Earthquake sequences are stochastic on two separated timescales: the seismic waves of any individual earthquake propagate over seconds to minutes, while the earthquakes themselves arrive as a sparse, self-exciting occurrence...

https://arxiv.org/abs/2610.38003

09 CONF

"It Would Make a Geordie Feel at Home": Regional Accent Perceptions and Speech Accommodation in Voice AI

ArXiv CS.AI 50%

arXiv:2610.38005v1 Announce Type: cross Abstract: Automatic speech recognition (ASR) systems often perform less accurately for speakers of regional dialects, but less is known about how users respond to these limitations and how regional voices produced by voice assistants (VAs)...

https://arxiv.org/abs/2610.38005

10 CONF

Measuring Deep Learning Performance - An Empirical Study of Performance Distributions Across Architectures and Tasks

ArXiv CS.AI 50%

arXiv:2610.38037v1 Announce Type: cross Abstract: Context: Non-determinism in deep learning algorithm design and implementation leads to performance variation, meaning model performance is not a single value, but rather a distribution. These model performance distributions are...

https://arxiv.org/abs/2610.38037

11 CONF

Prompt Lottery in LLM-as-a-Judge: Measuring and Mitigating Sensitivity to Syntactic Rephrasing

ArXiv CS.AI 50%

arXiv:2610.38076v1 Announce Type: cross Abstract: LLM-as-a-Judge has become a standard for evaluating large language models, yet most approaches implicitly assume that a single judge prompt functions as a stable evaluation instrument. Prior work has primarily explored deliberate...

https://arxiv.org/abs/2610.38076

12 CONF

Interpretable Feature Discovery from Unstructured Data Using LLMs and Statistical Choice Models

ArXiv CS.AI 50%

arXiv:2610.38082v1 Announce Type: cross Abstract: Feature extraction from unstructured data typically relies on opaque embeddings unsuited to interpretable decision-making. Interpretable features, on the other hand, typically require costly, and often infeasible,...

https://arxiv.org/abs/2610.38082

13 CONF

Jailbreak Connectivity: Towards Diverse, Transferable, and Universal Multimodal Large Language Model Jailbreak

ArXiv CS.AI 50%

arXiv:2610.38091v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) are vulnerable to visual jailbreak attacks that bypass safety safeguards. Existing methods typically optimize isolated adversarial examples, limiting attack diversity, transferability, and...

https://arxiv.org/abs/2610.38091

14 CONF

When Is Smaller Greener? Right-Sizing Language Models for Agentic AI

ArXiv CS.AI 50%

arXiv:2610.19345v2 Announce Type: replace Abstract: AI agents use a large language model (LLM) to plan and act over many steps. Which model should an agent use, so that its tasks are completed reliably with the least energy, cost and time? We treat this as a multi-criteria...

https://arxiv.org/abs/2610.19345

15 CONF

From the Gettier Machine to Justified True Output: A Three-Step Account of Rational Trust in Generative AI

ArXiv CS.AI 50%

arXiv:2610.38112v1 Announce Type: cross Abstract: As generative artificial intelligence (AI) settles into everyday practices of inquiry, an old epistemological question returns in a new guise: when does a proposition a user takes from an AI output become something the user...

https://arxiv.org/abs/2610.38112

16 CONF

Beauty Is Human: When People Love AI-Generated Art, They Think a Human Made It

ArXiv CS.AI 50%

arXiv:2610.38113v1 Announce Type: cross Abstract: Generative AI now produces images and poems that people cannot reliably distinguish from human-made work, so people often judge for themselves who made a work. We examine how these perceived-authorship judgments relate to...

https://arxiv.org/abs/2610.38113

17 CONF

Answer-Only Disclosure: Private-Context LLM Assistance over Web APIs with a Leakage Ledger

ArXiv CS.AI 50%

arXiv:2610.38126v1 Announce Type: cross Abstract: Web assistants increasingly answer questions about a user's private documents by pairing a small on-device model with a cloud LLM behind a Web API. In collaborative decoding, the document stays on the client, and the cloud...

https://arxiv.org/abs/2610.38126

18 CONF

Convex-SKM-Net: Low-Latency Block Halfspace Corrections for Neural Convex Constraint Satisfaction

ArXiv CS.AI 50%

arXiv:2610.38197v1 Announce Type: cross Abstract: Fast inference and strict constraint satisfaction are essential for learning-based decision-making. Existing methods often rely on constraint-specific projections or feasible-set parameterizations, while iterative repair can be...

https://arxiv.org/abs/2610.38197

19 CONF

Is it my preference? Role Grounding Attack against Agent's Memory via Social Interaction

ArXiv CS.AI 50%

arXiv:2610.38209v1 Announce Type: cross Abstract: LLM-based agents now share experiences on social platforms and remember new preferences to enhance their performance. However, existing memory systems contain a vulnerability in role-to-principal attribution that attackers can...

https://arxiv.org/abs/2610.38209

20 CONF

The Indian Financial Automated Advice Bias (IFAAB) Dataset : Auditing Socioeconomic and Demographic Disparities in LLM-Generated Financial Advice in India

ArXiv CS.AI 50%

arXiv:2610.38239v1 Announce Type: cross Abstract: Large language models are increasingly used for financial guidance, yet little is known about whether their advice changes across social identities in Global South contexts. We introduce the Indian Financial Automated Advice Bias...

https://arxiv.org/abs/2610.38239

21 CONF

Vendster: An Evaluation Arena to Investigate How Forecasting Tool Quality Affects LLM Agent Behaviors in Buyer-Vendor Negotiation

ArXiv CS.AI 50%

arXiv:2610.38292v1 Announce Type: cross Abstract: Time-series forecasters are ranked almost entirely by accuracy metrics, but a forecast's decision value, its effect on the downstream decision, need not track accuracy and is rarely measured. This problem compounds as large...

https://arxiv.org/abs/2610.38292

22 CONF

LLM-PowerTrace: A Dataset of Time-Resolved GPU Power Traces for AI Compute

ArXiv CS.AI 50%

arXiv:2610.38315v1 Announce Type: cross Abstract: AI compute is becoming large and fast-changing electrical loads. Its impacts on data center infrastructure and power grids depend not only on total energy consumption, but also on how their power draw moves over sub-second to...

https://arxiv.org/abs/2610.38315

23 CONF

Extremely Sparse-View Computed Tomography from 2D Projections via Pose-Aware Diffusion Priors

ArXiv CS.AI 50%

arXiv:2610.38319v1 Announce Type: cross Abstract: Extremely sparse-view computed tomography (xSV-CT) is critical when scanning time, motion, or radiation tolerance limits each scan to only a few projections. In many such scenarios, the same acquisition limits also constrain...

https://arxiv.org/abs/2610.38319

24 CONF

A Case Study of a Multi-Agent Kernel-Optimization Loop That Learns a New AI Accelerator

ArXiv CS.AI 50%

arXiv:2610.38329v1 Announce Type: cross Abstract: Frontier large language models write correct CUDA kernels for most single-operator benchmark tasks, yet their accuracy collapses on accelerators whose architecture and programming language they have rarely seen. We present a case...

https://arxiv.org/abs/2610.38329

25 CONF

LLMs as Judges and Examinees for Medical Item Difficulty Prediction

ArXiv CS.AI 50%

arXiv:2610.38402v1 Announce Type: cross Abstract: Predicting medical item difficulty from item text alone remains challenging. Prior work has largely relied on linguistic features or text representations derived from general-purpose and domain-specific language models, but...

https://arxiv.org/abs/2610.38402

26 CONF

SpaceAct: Benchmarking 3D Spatial and Temporal Reasoning via Closed-Loop Action and Active Perception

ArXiv CS.AI 50%

arXiv:2610.38405v1 Announce Type: cross Abstract: To solve problems in 3D space rather than merely describe images, multimodal large language models (MLLMs) need 3D spatiotemporal reasoning: reconstructing 3D geometry from occluded 2D views, anticipating physical dynamics, and...

https://arxiv.org/abs/2610.38405

27 CONF

The Code Behind the Layout: AI Agents for Analog Layout Generators

ArXiv CS.AI 50%

arXiv:2610.38426v1 Announce Type: cross Abstract: Analog layout remains largely manual and hard to automate. Generator frameworks such as gLayout express layouts as programs, producing layouts from parameterized, design-rule-aware generators; however, writing such generators...

https://arxiv.org/abs/2610.38426

28 CONF

Tracekit Box: Running AI Agents in Micro-VMs Where Every Way Out Is Decided and Signed

ArXiv CS.AI 50%

arXiv:2610.38458v1 Announce Type: cross Abstract: AI agents now install packages, call model APIs with their operators' keys and change code in repositories that deploy to production. The network, credentials and write access they need are also how they can leak data, steal a...

https://arxiv.org/abs/2610.38458

29 CONF

SaiVLA-JEPA: Evaluating Future Latent Prediction in JEPA-Conditioned VLA Control

ArXiv CS.AI 50%

arXiv:2610.38508v1 Announce Type: cross Abstract: Joint-embedding predictive architectures (JEPAs) learn to predict representations rather than pixels, offering a compact way to model how visual scenes evolve. We ask whether this predictive objective improves closed-loop control...

https://arxiv.org/abs/2610.38508

30 CONF

PACE: Pareto-Adaptive Compression for Efficient Native MLLMs

ArXiv CS.AI 50%

arXiv:2610.24005v2 Announce Type: replace-cross Abstract: Native multimodal large language models (native MLLMs) jointly encode visual and linguistic sequences within a unified autoregressive Transformer, emerging as a promising paradigm for multimodal understanding. Unlike...

https://arxiv.org/abs/2610.24005

31 CONF

Do Audio LLMs HEAR Pain? Tracing Acoustic Evidence from Speech to Language Reasoning

ArXiv CS.AI 50%

arXiv:2610.38570v1 Announce Type: cross Abstract: Large audio language models (LALMs) can process speech directly, but it remains unclear how well they use the non-lexical acoustic information that speech carries. We study this question through pain, using TAME-Pain, which...

https://arxiv.org/abs/2610.38570

32 CONF

Episodic looking - using reinforcement learning to bias perception enhances few-shot and continual learning with protection from forgetting

ArXiv CS.AI 50%

arXiv:2610.38577v1 Announce Type: cross Abstract: There is a fundamental contradiction between the desire for models that can generalize, and models that can continuously adapt to recent input with changing statistics. These conditions, captured in few-shot and continual...

https://arxiv.org/abs/2610.38577

33 CONF

How Deterministic SAST and AI Security Agents Divide the Work: Coverage, Harness Policy and Cost in Snyk VulnBench 2.0

ArXiv CS.AI 50%

arXiv:2610.38617v1 Announce Type: cross Abstract: We ask what AI coding agents and AI security harnesses find that deterministic static application security testing (SAST) does not, and what they miss that SAST catches. VulnBench 2.0 is a benchmark of 108 hand-curated,...

https://arxiv.org/abs/2610.38617

34 CONF

When AI performs, what remains? An integrative review of generative AI and cognitive capital in individuals and organizations

ArXiv CS.AI 50%

arXiv:2610.38647v1 Announce Type: cross Abstract: Generative artificial intelligence is rapidly entering knowledge work, and its short-term productivity benefits are increasingly documented. Less is known about what happens, over time, to the knowledge, skills and judgment that...

https://arxiv.org/abs/2610.38647

35 CONF

Information-Theoretic Reliability--Safety Bounds and Finite-Alignment Dynamics in AI Knowledge Distillation

ArXiv CS.AI 50%

arXiv:2610.38651v1 Announce Type: cross Abstract: The reliability and safety of artificial intelligence (AI), together with their transfer and evolution through knowledge distillation, have attracted growing attention. We establish an information-theoretic bound on the...

https://arxiv.org/abs/2610.38651

36 CONF

TrussTutor-Bench: Benchmarking large language models on solution accuracy, sycophancy, and error detection in statics

ArXiv CS.AI 50%

arXiv:2610.38917v1 Announce Type: cross Abstract: Large language models (LLMs) have rapidly become a default source of help for students in science and engineering courses. In foundational engineering mechanics courses such as statics, students routinely use these tools to check...

https://arxiv.org/abs/2610.38917

37 CONF

AtomTree: A Hierarchical Framework for State-Aware Embodied Instruction Following with LLMs

ArXiv CS.AI 50%

arXiv:2610.38744v1 Announce Type: cross Abstract: Embodied instruction following in interactive environments requires agents to interpret natural language commands and complete long-horizon tasks. Existing planners either rely on large labeled trajectories, or use few-shot and...

https://arxiv.org/abs/2610.38744

38 CONF

CopyLoop: Fast Looped Transformer Decoding via Context Copying and Shallow Screening

ArXiv CS.AI 50%

arXiv:2610.38822v1 Announce Type: cross Abstract: Looped Transformers increase computational capacity through the repeated application of shared blocks, but their recurrent depth makes conventional full-depth decoding expensive. While existing speculative pipelines accelerate...

https://arxiv.org/abs/2610.38822

39 CONF

From extreme weather to agricultural consequences with agentic AI

ArXiv CS.AI 50%

arXiv:2610.40641v1 Announce Type: cross Abstract: Extreme weather is forecast days ahead, yet its consequences for agriculture remain hard to predict: a forecast says where a storm will strike, not which crops will suffer or how much is at risk, and the records that could tell...

https://arxiv.org/abs/2610.40641

40 CONF

Interventional Training Reduces Causal Bypass in In-Context Introspection

ArXiv CS.AI 50%

arXiv:2610.38911v1 Announce Type: cross Abstract: Introspection, the ability of a system to access and reason about its internal signals, is crucial for language model (LM) self-consistency and scalable interpretability. Prior work argues that LMs possess introspective...

https://arxiv.org/abs/2610.38911

41 CONF

Benchmarking Graph Neural Networks for AC Power Flow: The Role of Inductive Biases and Physical Constraints

ArXiv CS.AI 50%

arXiv:2610.38985v1 Announce Type: cross Abstract: Fast and accurate models for AC power flow (ACPF) are increasingly important as modern electrical grids require more frequent system analyses. Graph neural networks (GNNs) naturally exploit electrical grid topology, yet the...

https://arxiv.org/abs/2610.38985

42 CONF

Implicit Reasoning in Transformers: Bridging the Compositionality Gap in GPT-2-like Models

ArXiv CS.AI 50%

arXiv:2610.39016v1 Announce Type: cross Abstract: Compositional generalization remains a persistent limitation of transformer language models: even when a model learns to combine known facts correctly in-distribution (ID), the same compositional mechanism may fail when applied...

https://arxiv.org/abs/2610.39016

43 CONF

Distilling the Shape of Uncertainty: What On-Policy Distillation Learns and How to Predict It

ArXiv CS.AI 50%

arXiv:2610.39020v1 Announce Type: cross Abstract: On-policy distillation (OPD) is increasingly adopted in the post-training of large language models. However, a stronger teacher may fail to improve the student. Prior work attributes this to a lack of new knowledge, but what that...

https://arxiv.org/abs/2610.39020

44 CONF

From Local Recovery to Unified Generation: Discovering Triple Set in Knowledge Graphs via Observation-Anchored Latent Diffusion

ArXiv CS.AI 50%

arXiv:2610.39114v1 Announce Type: cross Abstract: Triple set prediction (TSP) is to recover missing relational triples in a knowledge graph by predicting the triple set instead of isolated links. Existing approaches predominantly rely on the subgraph-based completion paradigm,...

https://arxiv.org/abs/2610.39114

45 CONF

Where Position Matters: Profiling-Guided Layer-Wise RoPE Allocation in Transformers

ArXiv CS.AI 50%

arXiv:2610.39223v1 Announce Type: cross Abstract: Rotary Position Embedding (RoPE) typically uses uniform coverage across Transformer layers despite heterogeneous positional requirements. Phase-off interventions reveal non-uniform, model-specific rotary-phase dependence, while...

https://arxiv.org/abs/2610.39223

46 CONF

There Is No Free Lunch in RLVR: Every pass@1 Gain Has a pass@k Price

ArXiv CS.AI 50%

arXiv:2610.39295v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards (RLVR) makes a language model solve more problems on the first try and fewer when given many tries: it raises pass@1 while the base model overtakes it at large sampling budget k. Is...

https://arxiv.org/abs/2610.39295

47 CONF

Learning to Bound: Monotonic, Calibrated Neural Upper Bounds for MPE Inference

ArXiv CS.AI 50%

arXiv:2610.39305v1 Announce Type: cross Abstract: Probabilistic graphical models (PGMs) compactly encode structured joint distributions, but Most Probable Explanation (MPE), which finds the highest-probability assignment to unobserved variables, remains computationally...

https://arxiv.org/abs/2610.39305

48 CONF

Grounded and Faithful P&ID Reasoning: Constraining Vision-Language Models with Recovered Evidence Graphs

ArXiv CS.AI 50%

arXiv:2609.05880v2 Announce Type: replace-cross Abstract: Piping and Instrumentation Diagrams (P&IDs) are the authoritative maps of process plants: isolation, maintenance, and HAZOP decisions depend on what connects to what. Vision-language models describe these sheets fluently,...

https://arxiv.org/abs/2609.05880

49 CONF

Cord: Confidence-Anchored Order Residual Decoding for Masked Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2610.39372v1 Announce Type: cross Abstract: The decoding order of masked diffusion language models (dLLMs) can substantially affect generation quality, yet common decoders typically apply the same reveal rule throughout generation. Confidence provides a strong default, but...

https://arxiv.org/abs/2610.39372

50 CONF

What Makes Medical Examination Items Difficult? Using Large Language Models to Explore Patterns Associated With Empirical Item Difficulty

ArXiv CS.AI 50%

arXiv:2610.39373v1 Announce Type: cross Abstract: Research on medical item difficulty has largely focused on prediction rather than on identifying interpretable patterns associated with empirical difficulty. This study used empirical difficulty labels to guide large language...

https://arxiv.org/abs/2610.39373

51 CONF

Learning to Formulate: Agentic Hardware-Software Co-Design of AI Infrastructure through Progressive Optimization and Verification

ArXiv CS.AI 50%

arXiv:2610.39380v1 Announce Type: cross Abstract: AI infrastructure, the chips and software that train and serve AI models, is designed in layers. Layers make the system manageable, but they also fix which choices an optimizer can consider. Gains that need the chip and its...

https://arxiv.org/abs/2610.39380

52 CONF

Sparse Fine-Tuning for Parameter-Efficient Adversarial Training

ArXiv CS.AI 50%

arXiv:2610.39455v1 Announce Type: cross Abstract: Deep neural networks achieve strong performance across many tasks but remain vulnerable to adversarial perturbations. Adversarial training (AT) is one of the most effective defenses, yet it suffers from high memory cost on large...

https://arxiv.org/abs/2610.39455

53 CONF

Integrating Heterogeneous Echocardiography AI Outputs in Language Space: Learnable Late Fusion with a Supervised LLM

ArXiv CS.AI 50%

arXiv:2610.39464v1 Announce Type: cross Abstract: View- and target-specific echocardiography AI models capture useful but complementary information and may not achieve sufficient predictive performance in isolation. A learned ensemble is a natural way to combine these outputs,...

https://arxiv.org/abs/2610.39464

54 CONF

LLM Agent-Based Design of RISC-V CPUs with FPGA-in-the-Loop Evaluation

ArXiv CS.AI 50%

arXiv:2610.39570v1 Announce Type: cross Abstract: Long-run RTL simulation of CPU candidates can take hours, while short diagnostic runs may miss performance degradation that emerges later in execution. This evaluation cost can delay design decisions in LLM-agent-based CPU...

https://arxiv.org/abs/2610.39570

55 CONF

Generative artificial intelligence prompting strategies among Chinese university students across academic tasks

ArXiv CS.AI 50%

arXiv:2610.39571v1 Announce Type: cross Abstract: Generative artificial intelligence (GenAI) is increasingly embedded in university learning, yet effective human-GenAI interaction depends not only on access to these tools but also on students' ability to formulate, evaluate, and...

https://arxiv.org/abs/2610.39571

56 CONF

Data Provenance and Ethics in the Curricula of U.S. Health AI Education Programs

ArXiv CS.AI 50%

arXiv:2610.39606v1 Announce Type: cross Abstract: Failures of deployed clinical artificial intelligence (AI) systems have often originated in how their training data were generated. Whether the programs that train the health AI workforce teach practitioners to examine data...

https://arxiv.org/abs/2610.39606

57 CONF

Test-Time Gradient Guidance of Flow Policies in Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2606.11087v2 Announce Type: replace-cross Abstract: Expressive continuous control policies, such as diffusion and flow models, form the backbone of recent advances in scaling imitation learning for simulated and real robot control. While they are known to scale stably in...

https://arxiv.org/abs/2606.11087

58 CONF

Multi-output LSTM soft sensors for industrial anaerobic co-digestion using LSTM-beta-VAE generative data augmentation

ArXiv CS.AI 50%

arXiv:2610.39768v1 Announce Type: cross Abstract: Anaerobic digestion (AD) requires close monitoring of key process variables, including pH, the ratio of volatile organic acids to alkaline buffer capacity and biogas production, to maintain biological stability and prevent...

https://arxiv.org/abs/2610.39768

59 CONF

SHPRR: An Algorithm for Pure Exploration under Constrained Regret

ArXiv CS.AI 50%

arXiv:2610.39837v1 Announce Type: cross Abstract: Best Arm Identification (BAI) conventionally measures the cost of experimentation by the number of samples collected. In many operational settings, however, the cost of exploration relative to a trusted baseline already in place...

https://arxiv.org/abs/2610.39837

60 CONF

Prompts and probabilities in the linguistic assessment of language models: Evidence from semantic focus-sensitivity

ArXiv CS.AI 50%

arXiv:2610.39923v1 Announce Type: cross Abstract: As language models (LMs) have improved, an increasingly common means of evaluating their linguistic competence is through explicit prompting (e.g., "Which sentence is more acceptable?"). Prior work indicates that such prompting...

https://arxiv.org/abs/2610.39923

61 CONF

Spectral Invariant Fusion and Refinement for Domain Generalized Semantic Segmentation

ArXiv CS.AI 50%

arXiv:2610.39926v1 Announce Type: cross Abstract: Domain generalized semantic segmentation is challenging because it requires accurate dense prediction on unseen domains without any access to target-domain data during training. Existing methods improve transferability through...

https://arxiv.org/abs/2610.39926

62 CONF

Mind The Gap (MTG-AD): Dataset and Fine-grained Evaluation for End-to-End Inline Audio Descriptions

ArXiv CS.AI 50%

arXiv:2610.39955v1 Announce Type: cross Abstract: Audio descriptions (AD) are essential for making visual media accessible to blind and low-vision (BLV) users. While Multimodal Large Language Models (MLLMs) offer a scalable solution for generating audio descriptions on-demand,...

https://arxiv.org/abs/2610.39955

63 CONF

Accumulating Pruning Constraints: New Exact Polynomial-Space Algorithms for Max 2-SAT

ArXiv CS.AI 50%

arXiv:2610.39992v1 Announce Type: cross Abstract: We present deterministic polynomial-space algorithms that solve Max 2-SAT on n variables exactly in time $\Ostar(2^{(1-\gamma)n})$, for a fixed $\gamma>0$, on several classes of instances whose clause density may grow with n....

https://arxiv.org/abs/2610.39992

64 CONF

UPPAAi: Towards LLM-Assisted Modeling, Verification, Analysis and Synthesis

ArXiv CS.AI 50%

arXiv:2610.40042v1 Announce Type: cross Abstract: For more than three decades, UPPAAL has supported the modelling and analysis of real-time and cyber-physical systems. Yet building a suitable model, formalising the intended properties, choosing analysis techniques and...

https://arxiv.org/abs/2610.40042

65 CONF

EPC-RawDC: Equivariant Physics Constrained 3D Diffusion with Raw Data Consistency for CBCT to CT Image Synthesis

ArXiv CS.AI 50%

arXiv:2610.40046v1 Announce Type: cross Abstract: Cone beam computed tomography (CBCT) is widely used for image guided and adaptive radiotherapy, but scatter, noise, reconstruction artifacts, and inaccurate Hounsfield units limit its quantitative use. Recent diffusion approaches...

https://arxiv.org/abs/2610.40046

66 CONF

One Vector or Many? Clustered Optimal-Transport Steering under Quantization

ArXiv CS.AI 50%

arXiv:2610.40055v1 Announce Type: cross Abstract: Activation steering changes what a language model generates by editing its hidden states at inference time. Its most common form adds one vector, the difference of class means (DiM), to every hidden state. CHaRS (concept...

https://arxiv.org/abs/2610.40055

67 CONF

Escaping Path Mirages in Offline Goal-Conditioned Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.40130v1 Announce Type: cross Abstract: Offline goal-conditioned reinforcement learning remains challenging in stochastic long-horizon settings, where compounding value estimation errors hinder reliable goal reaching. While prior methods have sought to address this...

https://arxiv.org/abs/2610.40130

68 CONF

FARE: Factor-Aware Retrieval with Error-Guided Fusion for Retail Forecasting

ArXiv CS.AI 50%

arXiv:2610.40160v1 Announce Type: cross Abstract: Retail sales forecasting underpins inventory planning, yet promotions and calendar events reshape short-term demand. Although each factor usually has historical occurrences, the complete factor combination of a target week often...

https://arxiv.org/abs/2610.40160

69 CONF

Self-training for cross-centre coronary vessel segmentation in X-ray angiography using pre-contrast frame subtraction

ArXiv CS.AI 50%

arXiv:2610.40302v1 Announce Type: cross Abstract: Deep learning models for coronary vessel segmentation in X-ray angiography often perform poorly on data from centres that were not represented in training. Apart from differences in equipment, the labels of the largest public...

https://arxiv.org/abs/2610.40302

70 CONF

When Context Changes: Understanding Update Failures in LLMs

ArXiv CS.AI 50%

arXiv:2609.38866v2 Announce Type: replace Abstract: As preferences, goals, and facts change, LLM agents must use the current state while earlier versions remain in context. Yet they can answer with an old value of the same variable, a failure that we call stale binding. To study...

https://arxiv.org/abs/2609.38866

71 CONF

AI in the EGI e-infrastructure -- Past experiences and future strategies

ArXiv CS.AI 50%

arXiv:2610.40494v1 Announce Type: cross Abstract: The EGI Federation has provided federated e-infrastructure services to European and global science for over 15 years. We review EGI's early experiences with AI in the interTwin and iMagine Horizon Europe projects, which built...

https://arxiv.org/abs/2610.40494

72 CONF

How I Aligned Your Model: Scaling Up Multilingual Safety Training with Templates

ArXiv CS.AI 50%

arXiv:2610.40516v1 Announce Type: cross Abstract: Safety alignment for Large Language Models is heavily focused on English with limited multilingual representation, particularly damaging low-resource language performance. Related open training datasets are often limited in size...

https://arxiv.org/abs/2610.40516

73 CONF

TextSeal2: Distortion-Free LLM Watermarking with Copula Mixing and Memoryless Races

ArXiv CS.AI 50%

arXiv:2610.40613v1 Announce Type: cross Abstract: We present TextSeal2, a distortion-free watermark for large language models that keeps its signal under speculative decoding: tokens that replace rejected drafts remain watermarked with the same keys, and the acceptance rate is...

https://arxiv.org/abs/2610.40613

74 CONF

Dual Exam: Controlled Evaluation of Multimodal Agents and World Models

ArXiv CS.AI 50%

arXiv:2610.31066v2 Announce Type: replace Abstract: World Models (WMs) can support Multimodal Agents (MAs) through simulated experience, while Agent trajectories can inform World Model refinement. Evaluating this interaction requires separate evidence about action selection and...

https://arxiv.org/abs/2610.31066

75 CONF

PRISM: Festina Lente Proactivity -- Risk-Sensitive, Uncertainty-Aware Deliberation for Proactive Agents

ArXiv CS.AI 50%

arXiv:2602.01532v2 Announce Type: replace Abstract: Proactive agents must decide not only what to say but also whether and when to intervene. Many current systems rely on brittle heuristics or indiscriminate long reasoning, which offers little control over the benefit-burden...

https://arxiv.org/abs/2602.01532

76 CONF

PRISM: Pushing the Frontier of Deep Think via Process Reward Model-Guided Inference

ArXiv CS.AI 50%

arXiv:2603.02479v2 Announce Type: replace Abstract: DEEPTHINK methods improve reasoning by generating, refining, and aggregating populations of candidate solutions, which enables strong performance on complex mathematical and scientific tasks. However, existing frameworks can...

https://arxiv.org/abs/2603.02479

77 CONF

Training Natural Schemers: RL Can Amplify Oversight-Conditional Behavior

ArXiv CS.AI 50%

arXiv:2610.26551v2 Announce Type: replace Abstract: One of the central concerns for the safety of advanced AI is scheming: when a model strategically complies under oversight but pursues its own goals when unobserved. Current demonstrations, such as Anthropic's Alignment Faking...

https://arxiv.org/abs/2610.26551

78 CONF

Beyond the Dirac Delta: Mitigating Diversity Collapse in Reinforcement Fine-Tuning for Versatile Image Generation

ArXiv CS.AI 50%

arXiv:2601.12401v4 Announce Type: replace-cross Abstract: Reinforcement learning (RL) has emerged as a paradigm for fine-tuning large-scale generative models, such as diffusion and flow models, to align with complex human preferences and user-specified tasks. A fundamental...

https://arxiv.org/abs/2601.12401

79 CONF

LLM-Driven, Datasheet-Aware Automated Hardware Compatibility Verification for Early-Stage, Pre-Schematic Embedded System Design

ArXiv CS.AI 50%

arXiv:2608.25217v2 Announce Type: replace Abstract: We present an LLM-driven, datasheet-aware framework for early-stage hardware compatibility verification that identifies documentation-level interface incompatibilities based on hardware datasheets and high-level component...

https://arxiv.org/abs/2608.25217

80 CONF

Mechanistic discovery in grain boundary structure and deformation driven by agentic AI

ArXiv CS.AI 50%

arXiv:2610.28629v2 Announce Type: replace-cross Abstract: Agentic artificial intelligence (AI) systems hold immense potential for automating complex workflows in materials discovery and design, but extending these capabilities to domain-intensive mechanistic studies remains...

https://arxiv.org/abs/2610.28629

81 CONF

Heavy-Tailed Memory Traces in Long-Horizon Language Agents

ArXiv CS.AI 50%

arXiv:2610.00010v2 Announce Type: replace Abstract: Long-horizon language agents are becoming increasingly important for tasks that require persistent interaction, planning, and reasoning over extended trajectories. External memory plays a central role in these agents by...

https://arxiv.org/abs/2610.00010

82 CONF

When Do Causal World Models Help Modular LLM Agents

ArXiv CS.AI 50%

arXiv:2610.00012v2 Announce Type: replace Abstract: LLM agents increasingly operate in modular systems, where planning requires coordinating multiple functional components whose actions can causally change downstream states and behaviors. This makes intervention-aware prediction...

https://arxiv.org/abs/2610.00012

83 CONF

MintFlow: Minimal Trajectory Intervention for Constrained Flow Matching

ArXiv CS.AI 50%

arXiv:2610.02260v2 Announce Type: replace Abstract: Flow matching models excel at generative modeling, and many downstream applications require their samples to satisfy prescribed constraints, such as observed measurements and physical laws. However, existing constrained...

https://arxiv.org/abs/2610.02260

84 CONF

Reliability of LLM Judges for Evaluating Entity Alignment

ArXiv CS.AI 50%

arXiv:2610.09554v2 Announce Type: replace Abstract: Entity Alignment (EA) identifies equivalent entities across knowledge graphs and is critical for knowledge base integration and ontology merging. Evaluating EA systems at scale requires expensive expert annotation, making...

https://arxiv.org/abs/2610.09554

85 CONF

Hippo: Training-Free Agent Self-Evolution via Hippocampal Replay

ArXiv CS.AI 50%

arXiv:2610.24453v2 Announce Type: replace Abstract: Motivated by recursive self-improvement (RSI), we study how LLM agents can learn through external memory without updating model parameters. Extracting useful experience from a single attempt is difficult: flawed reasoning can...

https://arxiv.org/abs/2610.24453

86 CONF

Joint Detection of Fraud and Concept Drift inOnline Conversations with LLM-Assisted Judgment

ArXiv CS.AI 50%

arXiv:2505.07852v2 Announce Type: replace-cross Abstract: Detecting fake interactions in digital communication platforms remains a challenging and insufficiently addressed problem. These interactions may appear as harmless spam or escalate into sophisticated scam attempts,...

https://arxiv.org/abs/2505.07852

87 CONF

Trustworthy AI and Democracy: A Dual Taxonomy and Operational Framework of Democratic Risks and Contributions

ArXiv CS.AI 50%

arXiv:2505.13565v3 Announce Type: replace-cross Abstract: Artificial Intelligence (AI) can undermine and strengthen democratic governance. This paper introduces the AIRD/AIPD framework, a dual taxonomy and operational framework for analyzing AI's democratic risks and conditional...

https://arxiv.org/abs/2505.13565

88 CONF

Generalizable dose denoising using diffusion transformer framework for pencil beam scanning proton therapy

ArXiv CS.AI 50%

arXiv:2506.04467v2 Announce Type: replace-cross Abstract: Background and purpose: Intensity-modulated proton therapy (IMPT) enables conformal dose delivery for head and neck (H&N) cancers. However, its sensitivity to inter-fractional anatomical changes motivates online...

https://arxiv.org/abs/2506.04467

89 CONF

Rethinking Group Recommender Systems in the Era of Generative AI: From One-Shot Recommendations to Conversational Group Decision Support

ArXiv CS.AI 50%

arXiv:2507.00535v3 Announce Type: replace-cross Abstract: More than twenty-five years ago, early ideas emerged on how to design systems that provide recommendations to groups of users rather than individuals. Since then, a wide range of algorithmic approaches has been proposed...

https://arxiv.org/abs/2507.00535

90 CONF

Membership Inference Attacks on Recommender System: A Survey

ArXiv CS.AI 50%

arXiv:2509.11080v5 Announce Type: replace-cross Abstract: Recommender systems (RecSys) have been widely applied to various applications, including E-commerce, finance, healthcare, social media and have become increasingly influential in shaping user behavior and decision-making,...

https://arxiv.org/abs/2509.11080

91 CONF

NeuroBridge: Bio-Inspired Self-Supervised EEG-to-Image Decoding via Cognitive Priors and Bidirectional Semantic Alignment

ArXiv CS.AI 50%

arXiv:2511.06836v2 Announce Type: replace-cross Abstract: Visual neural decoding seeks to reconstruct or infer perceived visual stimuli from brain activity patterns, providing critical insights into human cognition and enabling transformative applications in brain-computer...

https://arxiv.org/abs/2511.06836

92 CONF

MedGemma vs GPT-4: Open-Source and Proprietary Zero-shot Medical Disease Classification from Images

ArXiv CS.AI 50%

arXiv:2512.23304v2 Announce Type: replace-cross Abstract: Multimodal Large Language Models (LLMs) introduce an emerging paradigm for medical imaging by interpreting scans through the lens of extensive clinical knowledge, offering a transformative approach to disease...

https://arxiv.org/abs/2512.23304

93 CONF

Jacobian Scopes: A Unified Geometric Framework for Token-Level LLM Attribution

ArXiv CS.AI 50%

arXiv:2601.16407v5 Announce Type: replace-cross Abstract: Large language models (LLMs) make next-token predictions based on clues present in their context, such as semantic descriptions and in-context examples. Yet, elucidating which prior tokens most strongly influence a given...

https://arxiv.org/abs/2601.16407

94 CONF

Industrialized Deception: The Collateral Effects of LLM-Generated Misinformation on Digital Ecosystems

ArXiv CS.AI 50%

arXiv:2601.21963v3 Announce Type: replace-cross Abstract: Generative AI and misinformation research have evolved since our 2024 survey. This paper presents an updated perspective, transitioning from literature review to practical countermeasures. We report on changes in the...

https://arxiv.org/abs/2601.21963

95 CONF

ClueTracer: Question-to-Vision Clue Tracing for Training-Free Hallucination Suppression in Multimodal Reasoning

ArXiv CS.AI 50%

arXiv:2602.02004v2 Announce Type: replace-cross Abstract: Large multimodal reasoning models solve challenging visual problems via explicit long-chain inference: they gather visual clues from images and decode clues into textual tokens. Yet this capability also increases...

https://arxiv.org/abs/2602.02004

96 CONF

The Verification Crisis: Expert Perceptions of GenAI Disinformation and the Case for Reproducible Provenance

ArXiv CS.AI 50%

arXiv:2602.02100v2 Announce Type: replace-cross Abstract: The growth of Generative Artificial Intelligence (GenAI) has shifted disinformation production from manual fabrication to automated, large-scale manipulation. This article presents findings from the first wave of a...

https://arxiv.org/abs/2602.02100

97 CONF

The System Hallucination Scale (SHS): A Human-Centered Instrument for Assessing Perceived Reliability of Large Language Models

ArXiv CS.AI 50%

arXiv:2603.09989v2 Announce Type: replace-cross Abstract: Large language models routinely produce factually incorrect or misleading outputs, yet conventional automated benchmarks overlook how users perceive reliability during interaction. We introduce the System Hallucination...

https://arxiv.org/abs/2603.09989

98 CONF

GenoBERT: A Language Model for Accurate Genotype Imputation

ArXiv CS.AI 50%

arXiv:2604.00058v2 Announce Type: replace-cross Abstract: Genotype imputation enables dense variant coverage for genome-wide association and risk-prediction studies, yet conventional reference-panel methods remain limited by ancestry bias and reduced rare-variant accuracy. We...

https://arxiv.org/abs/2604.00058

99 CONF

Incentives Shape How Humans Co-Create with Generative AI

ArXiv CS.AI 50%

arXiv:2604.03529v2 Announce Type: replace-cross Abstract: Generative AI is quickly becoming an integral part of people's everyday workflows. Early evidence has shown that while generative AI can increase individual-level productivity, it does so at the cost of collective...

https://arxiv.org/abs/2604.03529

00 CONF

AffordSim: A Scalable Data Generator and Benchmark for Affordance-Aware Robotic Manipulation

ArXiv CS.AI 50%

arXiv:2604.11674v4 Announce Type: replace-cross Abstract: Many everyday robot manipulation skills are affordance-dependent, with success determined by whether the robot contacts the functional object region required by the subsequent action. Current simulation data generators...

https://arxiv.org/abs/2604.11674

01 CONF

ASTOR: Multi-Task Code Reinforcement Learning via Utility-Driven Coordination

ArXiv CS.AI 50%

arXiv:2605.06111v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) with verifiable rewards has proven effective at post-training LLMs for coding, yet deploying separate task-specific specialists incurs costs that scale with the number of tasks, motivating a...

https://arxiv.org/abs/2605.06111

02 CONF

The Illusion of Power Capping in LLM Decode: A Phase-Aware Energy Characterisation Across Attention Architectures

ArXiv CS.AI 50%

arXiv:2605.11999v2 Announce Type: replace-cross Abstract: Power capping is a common GPU energy lever in LLM serving, and it appears to work: the cap is set, the GPU reports drawing less than it, and the energy budget is met. We show the cap does little for the phase that...

https://arxiv.org/abs/2605.11999

03 CONF

Focused Forcing: Content-Aware Per-Frame KV Selection for Efficient Autoregressive Video Diffusion

ArXiv CS.AI 50%

arXiv:2605.18346v2 Announce Type: replace-cross Abstract: Recent advances in autoregressive video diffusion have enabled sequential and streaming video generation. However, long-horizon generation requires increasingly large KV caches, making efficient compression without...

https://arxiv.org/abs/2605.18346

04 CONF

FLORO: A Multimodal Geospatial Foundation Model for Ecological Remote Sensing Across Sensors and Scales

ArXiv CS.AI 50%

arXiv:2605.28174v2 Announce Type: replace-cross Abstract: Foundation models offer a promising route to transferable remote sensing representations, but many current approaches depend on very large pretraining datasets and fixed sensor configurations, limiting their suitability...

https://arxiv.org/abs/2605.28174

05 CONF

Can I Take Another Dose? DoseBench: Evaluating LLM Decision-Making in OTC Dosing QA

ArXiv CS.AI 50%

arXiv:2606.04262v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used to answer everyday health questions online, including whether someone can take another dose of an over-the-counter (OTC) medication. Yet this common safety-relevant...

https://arxiv.org/abs/2606.04262

06 CONF

JANUS: Evaluating Goal-Conditioned Distortion in Factually Grounded LLM Communication

ArXiv CS.AI 50%

arXiv:2606.10852v2 Announce Type: replace-cross Abstract: LLM deception is often evaluated through fabricated claims, explicit lies, or strategic concealment. Yet misleading communication can also arise from selectively presenting true facts, for example, by omitting adverse...

https://arxiv.org/abs/2606.10852

07 CONF

Reinforcement learning to improve large language model-based automated code compliance systems

ArXiv CS.AI 50%

arXiv:2606.22402v2 Announce Type: replace-cross Abstract: Large language model (LLM)-based approaches for automated code compliance (ACC) of building regulations are prone to generating incorrect and hallucinated computer-processable rules. This paper introduces P4IR, a...

https://arxiv.org/abs/2606.22402

08 CONF

GUEST: Preliminary Guidelines for Using and Evaluating GenAI Tools to Support Systematic Literature Reviews

ArXiv CS.AI 50%

arXiv:2607.24991v2 Announce Type: replace-cross Abstract: Context: Generative AI (GenAI) and Large Language Models (LLMs) are increasingly used for academic tasks in software engineering and beyond, including systematic literature reviews (SLRs). However, whilst their...

https://arxiv.org/abs/2607.24991

09 CONF

How Do Developers Edit AI-Generated Code, and How Well Do LLMs Infer Their Edits? Insights from 53.6K Developer Edits

ArXiv CS.AI 50%

arXiv:2607.25130v3 Announce Type: replace-cross Abstract: Imperfections in AI-generated code require that software developers modify generations through manual code edits. However, little is known about how developers edit AI-generated code, and to what extent LLMs can infer...

https://arxiv.org/abs/2607.25130

10 CONF

Decodable but Not Actionable: Auditing the Representation-Behavior Gap in Frozen LLMs for Geometric Constraints

ArXiv CS.AI 50%

arXiv:2608.17843v2 Announce Type: replace-cross Abstract: Large language models (LLMs) perform well on structured reasoning tasks, but what they encode and whether encoded information guides their behavior remain unclear. We introduce a controlled auditing framework that uses...

https://arxiv.org/abs/2608.17843

11 CONF

Iterative Calibration and Learning: Mitigating Teacher Bias in LLM Distillation without Target-Domain Reward Feedback

ArXiv CS.AI 50%

arXiv:2609.17474v2 Announce Type: replace-cross Abstract: Large language model (LLM) distillation can propagate teacher bias under covariate shift when target reward feedback is unavailable. We propose Iterative Calibration and Learning (IterCaL), which alternates source-reward...

https://arxiv.org/abs/2609.17474

12 CONF

Adaptive hybrid coupling with operator inference, the overlapping Schwarz alternating method and reinforcement learning

ArXiv CS.AI 50%

arXiv:2609.17837v2 Announce Type: replace-cross Abstract: Hybrid domain decomposition methods provide a flexible framework for coupling full order models (FOMs) and reduced order models (ROMs), but typically assume the model assigned to each subdomain is fixed throughout a...

https://arxiv.org/abs/2609.17837